2.10 Domain, Range and Composite Functions

Worked Solutions 1–39

1 a)

A function as every point maps to a single other point

1 b)

Not a function as 6 has two outputs: both 3 and 2

1 c)

Not a function as 1 has two outputs: both -1 and 1

1 d)

A function as every point maps to a single other point

2 a)

Many-to-one, a function

2 b)

One-to-one, a function

2 c)

Many-to-one, not a function due to asymptote

2 d)

One-to-many, not a function

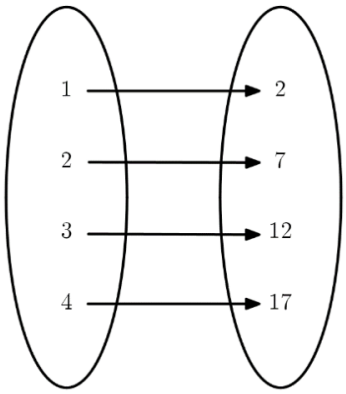
2 e)

Many-to-many, not a function

2 f)

Many-to-one, a function

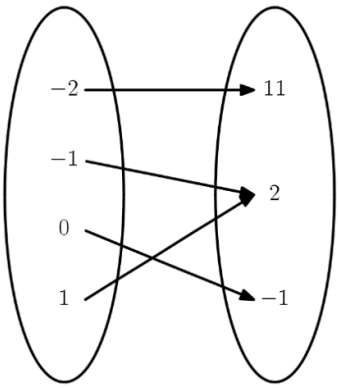
3 a)



3 b)

It is a one-to-one function.

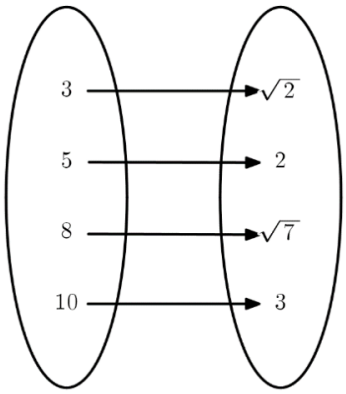
4 a)



4 b)

It is a function, many-to-one.

5 a)



5 b)

It is a function, one-to-one.

6 a)

One-to-one.

6 b)

Many-to-one.

6 c)

One-to-one.

6 d)

One-to-one.

6 e)

Many-to-one.

6 f)

One-to-one.

6 g)

One-to-one.

6 h)

Many-to-one.

6 i)

One-to-one.

6 j)

One-to-one.

7 a)

$\{f(x) : f(x) \in \mathbb{R}, f(x) > 1\}$.

7 b)

$\{y : y \in \mathbb{R}, y \geq -5\}$.

7 c)

$\{f(x) : f(x) \in \mathbb{R}\}$.

7 d)

$\{y : y \in \mathbb{R}, y < 0\}$.

7 e)

$\{f(x) : f(x) \in \mathbb{R}, f(x) < 5\}$.

8 a)

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{y : y \in \mathbb{R}, -1 \leq y \leq 1\}$.

8 b)

Domain: $\{x : x \in \mathbb{R}, x \geq 0\}$.

Range: $\{y : y \in \mathbb{R}, y \geq 0\}$.

8 c)

Maximum at $(-4, -9)$.

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{y : y \in \mathbb{R}, y \leq -9\}$.

8 d)

Minimum at $(\frac{3}{2}, \frac{9}{2})$.

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{y : y \in \mathbb{R}, y \geq \frac{9}{2}\}$.

8 e)

Domain: $\{x : x \in \mathbb{R}, x \neq 3\}$.

Range: $\{y : y \in \mathbb{R}, y \neq 0\}$.

8 f)

Domain: $\{x : x \in \mathbb{R}, x \neq 0\}$.

Range: $\{y : y \in \mathbb{R}, y \neq 4\}$.

8 g)

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{y : y \in \mathbb{R}, y > 2\}$.

8 h)

Domain: $\{x : x \in \mathbb{R}, x > 0\}$.

Range: $\{y : y \in \mathbb{R}\}$.

8 i)

Domain: $\{x : x \in \mathbb{R}, x > 2\}$.

Range: $\{y : y \in \mathbb{R}\}$.

9 a)

Domain: $\{x : x \in \mathbb{R}, -5 \leq x \leq 3\}$.

Range: $\{y : y \in \mathbb{R}, -2 \leq y \leq 6\}$.

9 b)

Domain: $\{x : x \in \mathbb{R}, -3 < x < 4\}$.

Range: $\{y : y \in \mathbb{R}, -5 < y < 9\}$.

9 c)

Domain: $\{x : x \in \mathbb{R}, -2 \leq x < 2\}$.

Range: $\{y : y \in \mathbb{R}, -8 \leq y < 8\}$.

9 d)

Domain: $\{x : x \in \mathbb{R}, -3 \leq x < 2\}$.

Range: $\{y : y \in \mathbb{R}, 0 \leq y \leq 9\}$.

9 e)

Domain: $\{x : x \in \mathbb{R}, -3 \leq x \leq 4\}$.

Range: $\{y : y \in \mathbb{R}, -\sqrt{7} \leq y \leq \sqrt{7}\}$.

9 f)

Domain: $\{x : x \in \mathbb{R}, -2 \leq x < 3\}$.

Range: $\{y : y \in \mathbb{R}, -6.5 < y \leq 7\}$.

9 g)

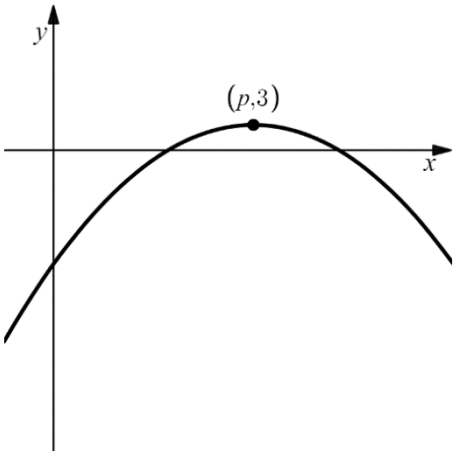
Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{y : y \in \mathbb{R}, -2\pi < y < 2\pi\}$.

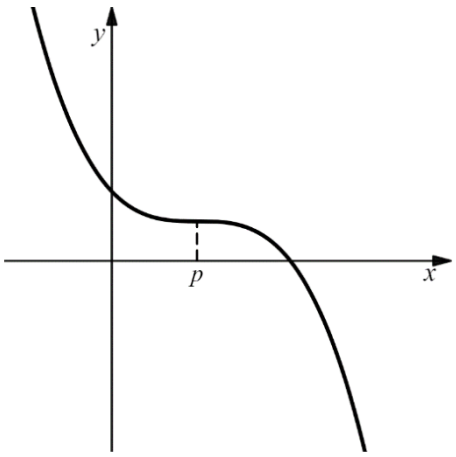
9 h)

Domain: $\{x : x \in \mathbb{R}, -\pi \leq x < -\frac{\pi}{2}\} \cup \{x : x \in \mathbb{R}, -\frac{\pi}{2} < x < \frac{\pi}{2}\} \cup \{x : x \in \mathbb{R}, \frac{\pi}{2} < x \leq \pi\}$. Range: $\{y : y \in \mathbb{R}, y \geq 4\} \cup \{y : y \in \mathbb{R}, y \leq 0\}$.

10)



11)



12 a)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq 0\}$.

12 b)

$f(2) = \frac{1}{2}(2-3)^2 - 2 = -\frac{3}{2}$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -\frac{3}{2}\}$.

12 c)

$f(0) = \frac{1}{2}(0-3)^2 - 2 = \frac{5}{2}$. Minimum at $(3, -2)$.

Range: $\{f(x) : f(x) \in \mathbb{R}, -2 \leq f(x) \leq \frac{5}{2}\}$.

13 a)

$-0.5x^2 + 5x - 12.5 = -0.5(x-5)^2 + 18$. Maximum at $(5, 18)$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq 18\}$.

13 b)

$f(2) = -0.5 \times 2^2 + 5 \times 2 + 5.5 = 13.5.$

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq 13.5\}$.

13 c)

$f(0) = 5.5$. $f(4) = -0.5 \times 4^2 + 5 \times 4 + 5.5 = 17.5.$

Range: $\{f(x) : f(x) \in \mathbb{R}, 5.5 < f(x) < 17.5\}$.

13 d)

$f(3) = -0.5 \times 3^2 + 5 \times 3 + 5.5 = 16 = f(7).$

Range: $\{f(x) : f(x) \in \mathbb{R}, 16 \leq f(x) \leq 18\}$.

14 a)

$f(3) = \frac{1}{3}(3-5)^2 e^3 = \frac{1}{3}e^3.$

Range: $\{f(x) : f(x) \in \mathbb{R}, 0 < f(x) \leq \frac{1}{3}e^3\}$.

14 b)

Range: $\{f(x) : f(x) \in \mathbb{R}, 0 \leq f(x) \leq \frac{1}{3}e^3\}$.

14 c)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq 0\}$.

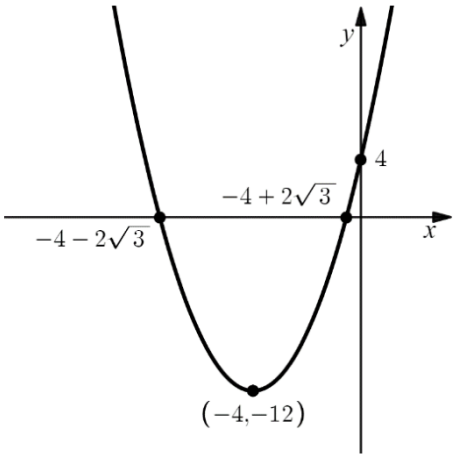
14 d)

$f(0) = \frac{1}{12}(0-5)^2 e^0 = \frac{25}{12}.$

Range: $\{f(x) : f(x) \in \mathbb{R}, \frac{25}{12} \leq f(x) < \frac{1}{3}e^3\}$.

15 a)

$f(x) = (x+4)^2 - 12$. Minimum at $(-4, -12)$. $x^2 + 8x + 4 = 0 \Rightarrow x = -4 \pm 2\sqrt{3}$.



15 b)

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -12\}$.

16 a)

$f(1) = 4 + 1 = 5$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq 5\}$.

16 b)

$f(-2) = \frac{1}{3} \times -2 - 2 = -\frac{8}{3}$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) < -\frac{8}{3}\}$.

16 c)

$f(-2) = 5 - 8 \times -2 = 21$. $f(4) = 5 - 8 \times 4 = -27$.

Range: $\{f(x) : f(x) \in \mathbb{R}, -27 < f(x) \leq 21\}$.

16 d)

$f(4) = 2\sqrt{4} = 4$. $f(16) = 2\sqrt{16} = 8$.

Range: $\{f(x) : f(x) \in \mathbb{R}, 4 < f(x) \leq 8\}$.

16 e)

$f(0) = 0 = f(3)$. Maximum point at $(1.5, 2.25)$.

Range: $\{f(x) : f(x) \in \mathbb{R}, 0 < f(x) \leq 2.25\}$.

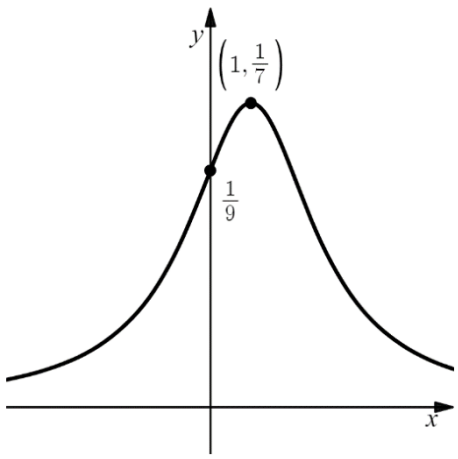
16 f)

Max when x^2 minimised, that is when $x = 0$. $f(0) = 1$.

Range: $\{f(x) : f(x) \in \mathbb{R}, 0 < f(x) \leq 1\}$.

17)

$f(x) = \dfrac{1}{2x^2 - 4x + 9} = \dfrac{1}{2(x-1)^2 + 7}$. Max when $(x-1)^2$ minimised, so max is $f(1) = \frac{1}{7}$. $2(x-1)^2 + 7 \geq 0$ so $f(x) > 0$ for all x.



Range: $\{f(x) : f(x) \in \mathbb{R}, 0 < f(x) \leq \frac{1}{7}\}$.

18)

$$\begin{aligned} 2x^2 + kx + 13 &= 2\left[x^2 + \frac{1}{2}k\right] + 13 \\ &= 2\left(x + \frac{1}{4}k\right)^2 - \frac{2}{16}k^2 + 13. \end{aligned}$$

$-\frac{1}{8}k^2 + 13 = -5 \Rightarrow k^2 = 144 \Rightarrow k = \pm 12.$

19)

To be one-to-one p must be the x-value of the minimum.

$$\begin{aligned} k(x-a)(x-b) &= k[x^2 - (a+b)x + ab] \\ &= k\left[\left(x - \frac{1}{2}(a+b)\right)^2 - \frac{1}{4}(a+b)^2 + ab\right]. \end{aligned}$$

Minimum when $x = \frac{1}{2}(a+b)$. $p = \frac{1}{2}(a+b)$.

20 a)

$f'(x) = -4x^3 + 4$. Stat pt when $f'(x) = 0$. $-4x^3 + 4 = 0 \Rightarrow x^3 = 1 \Rightarrow x = 1$. $f(1) = -1 + 4 = 3$.

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq 3\}$.

20 b)

$f'(x) = 4 - 2e^{2x}$. Stat pt when $f'(x) = 0$. $4 - 2e^{2x} = 0 \Rightarrow e^{2x} = 2 \Rightarrow 2x = \ln 2 \Rightarrow x = \frac{1}{2}\ln 2$. $f(\frac{1}{2}\ln 2) = 4 \times \frac{1}{2}\ln 2 - e^{2\times\frac{1}{2}\ln 2} = 2\ln(2) - 2$.

Domain: $\{x : x \in \mathbb{R}\}$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq 2\ln(2) - 2\}$.

21)

$f'(x) = 1 + \ln x$. Stat pt when $f'(x) = 0$. $1 + \ln x = 0 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1}$. $f(e^{-1}) = (e^{-1})\ln(e^{-1}) = -\frac{1}{e}$.

Domain: $\{x : x \in \mathbb{R}, x \geq 0\}$.

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -\frac{1}{e}\}$.

22 a)

$f(-2) = (-2)^3 - 3(-2)^2 - 24(-2) = 28$. $f'(x) = 3x^2 - 6x - 24$. Stat point when $f'(x) = 0 \Rightarrow 3x^2 - 6x - 24 = 0$. $x = 4, -2$. $f(4) = -80$ so minimum at $(4, -80)$.

Range: $\{f(x) : f(x) \in \mathbb{R}, -80 \leq f(x) < 28\}$.

22 b)

Range: $\{g(x) : g(x) \in \mathbb{R}, -160 \leq g(x) < 56\}$.

22 c)

Range: $\{h(x) : h(x) \in \mathbb{R}, -244 \leq h(x) < 80\}$.

23 a)

$fg(-1) = f((-1)^2 + 2) = f(3) = 15 - 1 = 14.$

23 b)

$gf(2) = g(10 - 1) = g(9) = 81 + 2 = 83.$

23 c)

$ff(4) = f(20 - 1) = f(19) = 95 - 1 = 94.$

24)

$g(4) = \frac{1}{4-2} = \frac{1}{2}$. $fg(4) = f(\frac{1}{2}) = \frac{1}{4}$.

25 a)

$fg(x) = f(x^2) = 2(x^2) + 1 = 2x^2 + 1.$

25 b)

$gf(x) = g(2x + 1) = (2x + 1)^2.$

25 c)

$ff(x) = f(2x + 1) = 2(2x + 1) + 1 = 4x + 3.$

25 d)

$fg(a) = 2a^2 + 1 = 33 \Rightarrow a^2 = 16$. a is positive $\therefore a = 4$.

25 e)

$2x^2 + 1 = (2x + 1)^2$. $2x^2 + 1 = 4x^2 + 4x + 1$. $2x^2 + 4x = 0 \Rightarrow 2x(x + 2) = 0 \Rightarrow x = 0$ or $x = -2$.

26 a)

$f(x) = \frac{1}{x}$, $ff(x) = \frac{1}{1/x} = x$, $fff(x) = \frac{1}{x}$, ...

$f^{89}(x) = \frac{1}{x}$.

26 b)

$f^{100}(x) = x.$

27)

$$gf(x) = g(1 + \sqrt{x}) = 3 - 2^{-(1+\sqrt{x})},$$
$$3 - 2^{-(1+\sqrt{x})} = \frac{23}{8} \Rightarrow -2^{-(1+\sqrt{x})} = -\frac{1}{8} \Rightarrow 2^{-(1+\sqrt{x})} = \frac{1}{8},$$
$$2^{1+\sqrt{x}} = 8 \Rightarrow 1 + \sqrt{x} = 3 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4.$$

28 a)

$fg(x) = f(10^x) = 5\log_{10}(10^x) = 5x.$

28 b)

$$\begin{aligned} fgf(x) = fg(5\log_{10} x) &= f(10^{5\log_{10} x}) = f(10^{\log_{10} x^5}) \\ &= f(x^5) = 5\log_{10} x^5 = 25\log_{10} x. \end{aligned}$$

29)

$f(2) = -4k + 24 - 11 = -4k + 13.$

$$\begin{aligned} f(f(2)) &= f(-4k+13) \\ &= -k(-4k+13)^2 + 12(-4k+13) - 11 \\ &= -k(16k^2 - 104k + 169) - 48k + 156 - 11 \\ &= -16k^3 + 104k^2 - 169k - 48k + 145 \\ &= -16k^3 + 104k^2 - 217k + 145. \end{aligned}$$

$ff(2) = -2 \Rightarrow -16k^3 + 104k^2 - 217k + 145 = -2.$ $-16k^3 + 104k^2 - 217k + 147 = 0 \Rightarrow k = 3$ or $k = \frac{7}{4}$.

30 a)

$-3.$

30 b)

$ff(\frac{1}{2}) = f(\frac{1}{2} + 4) = f(\frac{9}{2}) = 7 - 2 \times \frac{9}{2} = -2.$

30 c)

$x + 4 = -2 \Rightarrow x = -6.$ $7 - 2x = -2 \Rightarrow x = \frac{9}{2}.$

$\{x : x \in \mathbb{R}, x < -6\} \cup \{x : x \in \mathbb{R}, x > \frac{9}{2}\}.$

31 a)

1.

31 b)

$\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -4\}.$

31 c)

$ff(2) = f(4 - 4 - 3) = f(-3) = -(-3) - 3 = 0.$

31 d)

$-x - 3 = 0 \Rightarrow x = -3.$ $x^2 - 2x - 3 = 0 \Rightarrow x = 3$ or $x = -1 \Rightarrow x = 3.$

$\{x : x \in \mathbb{R}, -3 < x < 3\}.$

32 a)

$f(a) = 5 \Rightarrow 5 = 6 - e^{a-4} \Rightarrow 1 = e^{a-4} \Rightarrow \ln 1 = a - 4 \Rightarrow a = 4.$

32 b)

$ff(\frac{7}{2}) = f(\frac{7}{2} + 1) = f(\frac{9}{2}) = 6 - e^{\frac{9}{2}-4} = 6 - e^{1/2}.$

32 c)

$x + 1 = 0 \Rightarrow x = -1.\ 6 - e^{x-4} = 0 \Rightarrow 6 = e^{x-4} \Rightarrow \ln 6 = x - 4 \Rightarrow x = 4 + \ln 6.$

$\{x : x \in \mathbb{R}, -1 \leq x \leq 4 + \ln 6\}.$

33 a)

	2	
x	$2x$	$+11$
-5	-10	

$\dfrac{2x+1}{x-5} = 2 + \dfrac{11}{x-5}.$

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \neq 2\}.$

33 b)

$$ff(x) = f\left(\frac{2x+1}{x-5}\right) = \frac{2\left(\frac{2x+1}{x-5}\right)+1}{\left(\frac{2x+1}{x-5}\right)-5} = \frac{2(2x+1)+(x-5)}{(2x+1)-5(x-5)}$$
$$= \frac{4x+2+x-5}{2x+1-5x+25} = \frac{5x-3}{26-3x}.$$

34 a)

	a	
x	ax	$4a+1$
-4	$-4a$	

$\dfrac{ax+1}{x-4} = a + \dfrac{4a+1}{x-4}.$

Range: $\{f(x) : f(x) \in \mathbb{R}, f(x) \neq a\}.$

34 b)

$$ff(x) = f\left(\frac{ax+1}{x-4}\right) = \frac{a\left(\frac{ax+1}{x-4}\right)+1}{\left(\frac{ax+1}{x-4}\right)-4} = \frac{a(ax+1)+(x-4)}{(ax+1)-4(x-4)}$$
$$= \frac{a^2x+a+x-4}{ax+1-4x+16} = \frac{(a^2+1)x+a-4}{(a-4)x+17}.$$

35 a)

$fg(x) = f(a^2 - x) = (a^2 - x)^2 - a(a^2 - x)$.

35 b)

$$\begin{aligned} fg(2) = f(a^2 - 2) &= (a^2 - 2)^2 - a(a^2 - 2) \\ &= (a^2 - 2)(a^2 - 2 - a) \\ &= (a^2 - 2)(a - 2)(a + 1) \\ &= (a^2 - 2)(a - 2)(a + 1) = 0. \end{aligned}$$

$a = 2$, $a = -1$, $a = \sqrt{2}$, $a = -\sqrt{2}$.

36 a)

$f(-3) = 0$. Range: $f(x) \in (0, \infty)$.

36 b)

$f(-1) = 2(-1 + 3)^2 e^{-1} = 8e$. Range: $f(x) \in [0, 8e]$.

36 c)

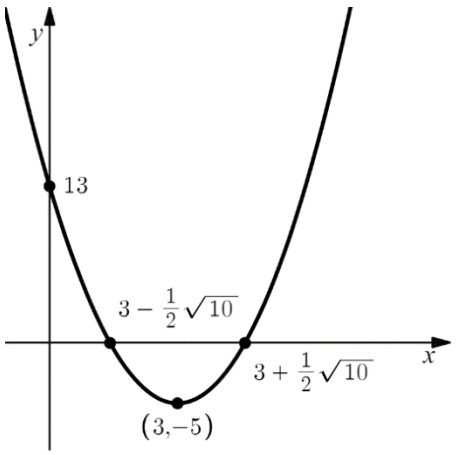
$f(0) = 2(0 + 3)^2 e^0 = 18$. Range: $f(x) \in (0, 18]$.

36 d)

Range: $f(x) \in [0, 8e]$.

37 a)

$f(x) = 2(x - 3)^2 - 5$. Minimum at $(3, -5)$. $2x^2 - 12x + 13 = 0 \Rightarrow x = 3 \pm \frac{1}{2}\sqrt{10}$.



37 b)

Range: $f(x) \in [-5, \infty)$.

38 a)

$f'(x) = 80x^4 - 45x^2 = 5x^2(16x^2 - 9)$. Stat point when $f'(x) = 0 \Rightarrow x^2 = 0$ or $16x^2 - 9 = 0$. $x = 0, -\frac{3}{4}, \frac{3}{4}$.

$$f(-\tfrac{3}{4}) = 16(-\tfrac{3}{4})^5 - 15(-\tfrac{3}{4})^3 + 1 = -\frac{243}{64} + \frac{405}{64} + 1 = \frac{113}{32},$$
$$f(\tfrac{3}{4}) = 16(\tfrac{3}{4})^5 - 15(\tfrac{3}{4})^3 + 1 = \frac{243}{64} - \frac{405}{64} + 1 = -\frac{49}{32}.$$

So maximum at $(-\frac{3}{4}, \frac{113}{32})$ and minimum at $(\frac{3}{4}, -\frac{49}{32})$. Range: $f(x) \in [-\frac{49}{32}, \frac{113}{32}]$.

38 b)

Range: $f(x) \in [-98, 226]$.

38 c)

Range: $f(x) \in [-\frac{89}{4}, \frac{73}{4}]$.

38 d)

Domain: $x \in (-\frac{1}{2}, \frac{1}{2})$.

39 a)

$f(a) = -1 \Rightarrow -1 = e^{a-2} - 2 \Rightarrow 1 = e^{a-2} \Rightarrow \ln 1 = a - 2 \Rightarrow a = 2$.

39 b)

$f(2) = -1 \Rightarrow 3 - 2k = -1 \Rightarrow 4 = 2k \Rightarrow k = 2$.

39 c)

$ff(3) = f(e^{3-2} - 2) = f(e - 2) = 3 - 2(e - 2) = 3 - 2e + 4 = 7 - 2e$.

39 d)

$3 - 2x = 0 \Rightarrow x = \frac{3}{2}$. $e^{x-2} - 2 = 0 \Rightarrow e^{x-2} = 2 \Rightarrow x - 2 = \ln 2 \Rightarrow x = 2 + \ln 2$.

$x \in [\frac{3}{2}, 2 + \ln 2]$.